Evaluating the ability of a behavioral foundation model to predict human neuromechanics

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MGAIC brain+cognitive sciences
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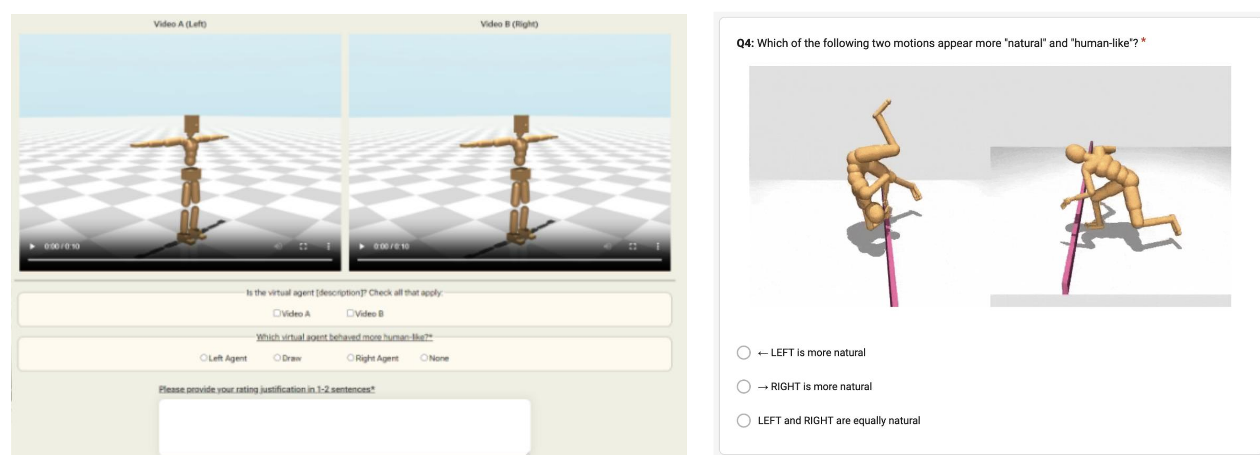
Behavioral foundation models (BFMs)

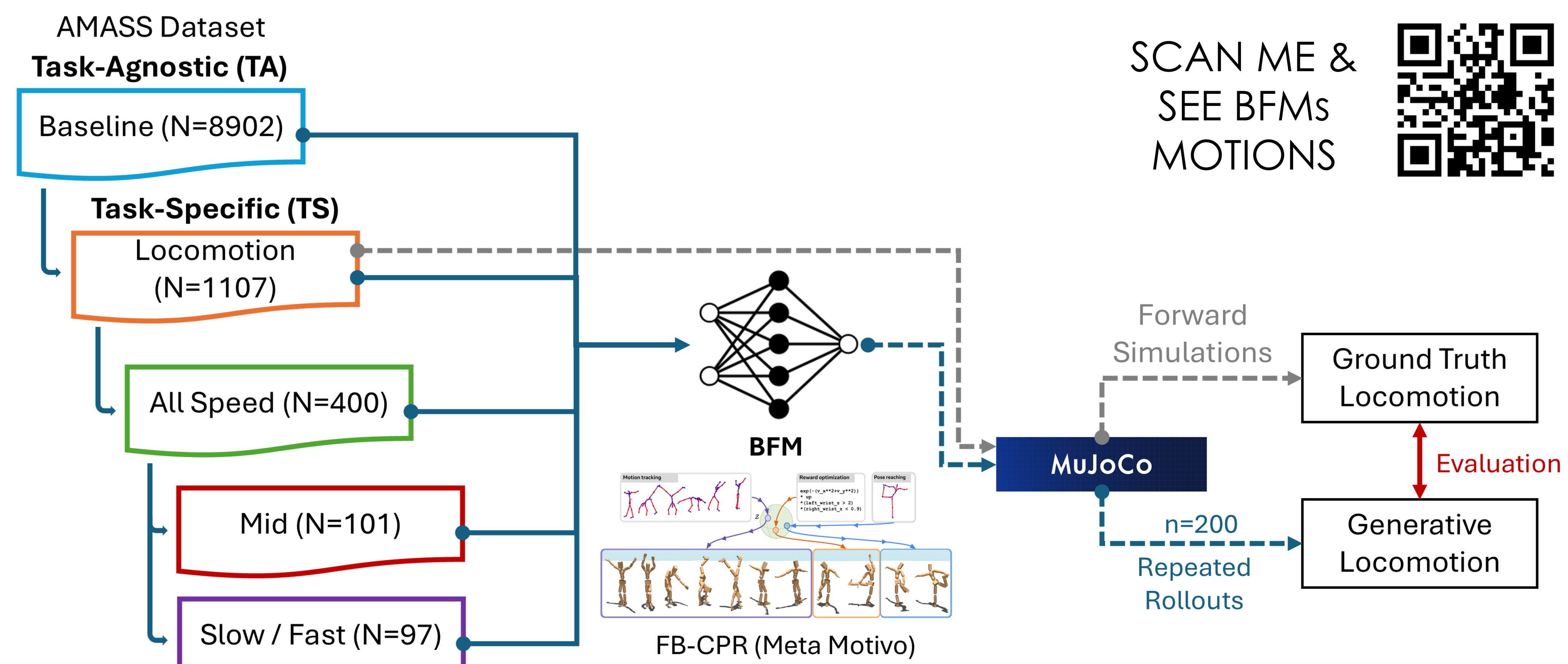
- Task-general generative AI for humanoids [1]
- Existing evaluation metrics are based on visually assessed human-likeness [1, 2, 3]



User surveys for human-likeness score [1, 2]

Can BFMs capture human-like neuromechanics?

Human-likeness evaluation framework



We aim to assess the **human-likeness of behavioral foundation models** with neuromechanical phenomena.

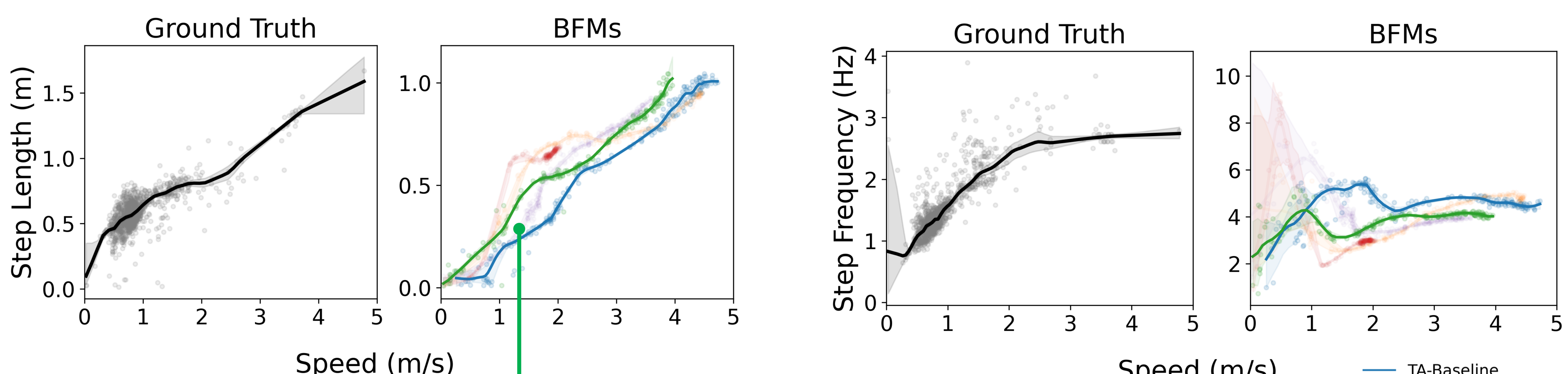
Key takeaways

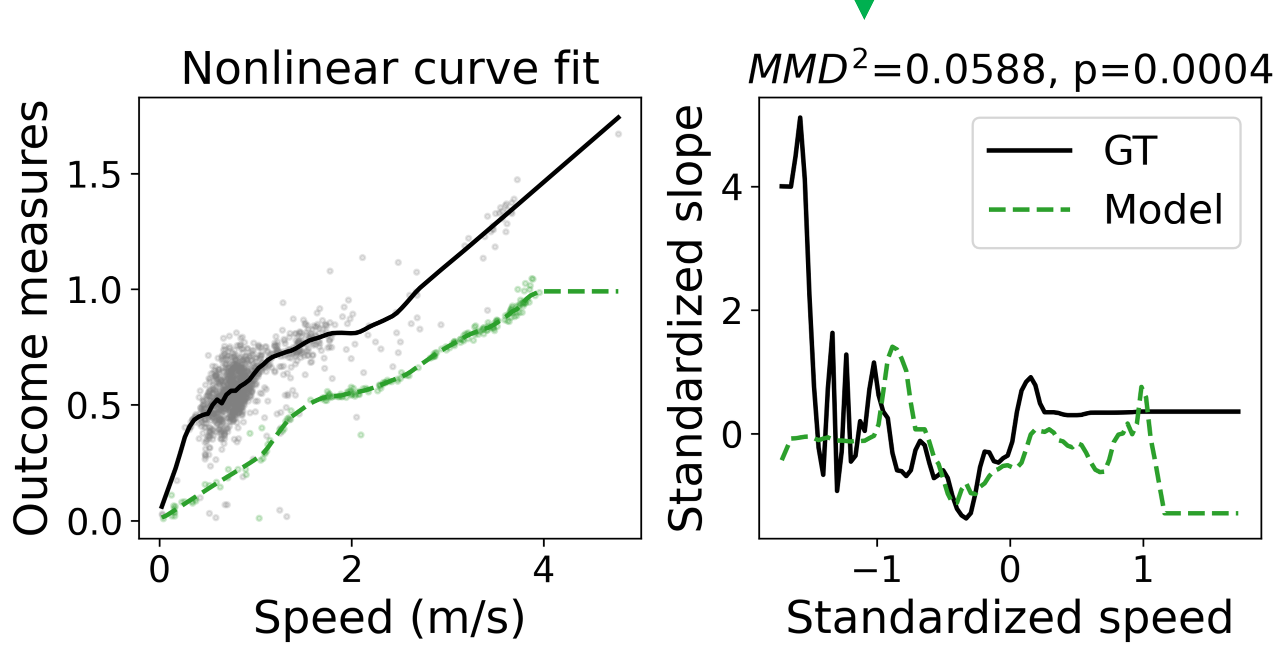
- Importance of **neuromechanics-aware** evaluations of behavioral foundation models.

Limitations and future work

- Evaluation against other neuromechanical phenomena like joint dynamics and ground reaction forces.
- Integrate neuromechanical metrics into the training.
- End-to-end benchmarking that is generalizable across other BFM platforms and algorithms

#1 Sample inefficiency when capturing classic locomotor relationships



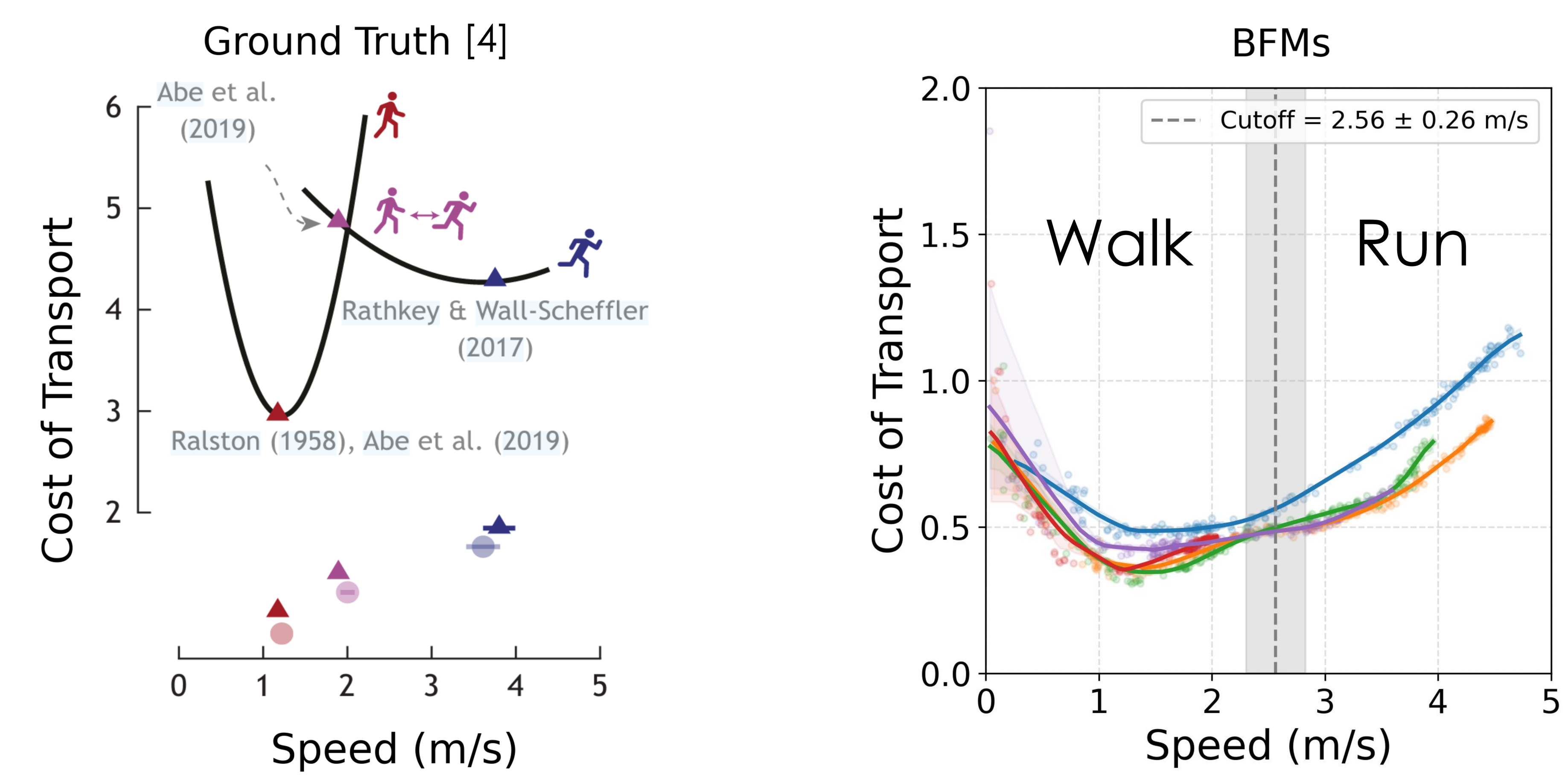


Slope Similarity using Squared Max Mean Discrepancy (MMD²)

	Speed - Step Len	Spd - Step Freq
TA-Baseline	**0.0411**	0.0358
TS-Large-All	0.0747	0.0325
TS-Small-All	0.0588	**0.0255**
TS-Small-Mid	0.1522	0.0454
TS-Small-slow/fast	0.0754	0.0691

- **Task-specific BFMs** captured human-like locomotion comparably to task-agnostic BFMs.
- Task-specific BFMs could be a **sample-efficient** alternative to task-agnostic BFMs.

#2 Absence of human-like energy landscape for gait transition



$$\text{Cost of Transport} = \frac{P}{m \cdot g \cdot v}$$

- **Walk-run energy landscape** that theoretically determines human gait transitions was not captured by the BFMs.

References
[1] Tirinzoni, Andrea, et al. "Zero-shot whole-body humanoid control via behavioral foundation models." arXiv preprint arXiv:2504.11054 (2025).
[2] Hansen, Nicklas, et al. "Hierarchical world models as visual whole-body humanoid controllers." arXiv preprint arXiv:2405.18418 (2024).
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[4] McAllister, Megan J., Anthony Chen, and Jessica C. Selinger. "Behavioural energetics in human locomotion: how energy use influences how we move." Journal of Experimental Biology 228.Suppl_1 (2025): JEB248125.